Machine Learning Project Requirements

The project must be completed in groups, with a maximum of 5 students per group

1. **Problem Formulation & Data Sourcing**

- The Problem: Define the practical problem, the target stakeholder, the Target Variable, and the task type (Binary/Multi-class Classification, or Regression). Identify the most costly error type (False Positives vs. False Negatives).
- Data Sources: Raw Kaggle/UCI datasets are strictly prohibited. You must create a dataset (>=1,000 rows) by: merging multiple sources.

  Fabricating data is forbidden.

2. **EDA, Splitting & Preprocessing**

- EDA & Data Dictionary: Analyze distributions, missing values, duplicates, and class imbalance. Submit a Data Dictionary detailing each column's type, meaning, source, and processing method.
- Data Leakage Prevention: Split the data (Train/Test) *before* any processing. Use stratified splitting for imbalanced datasets.
- Cleaning & Pipelines: Handle missing values, outliers, and encoding. Imputation, scaling, and encoding must be executed within a Pipeline or fitted solely on training data.
- Scientific Justification: Every cleaning decision (e.g., median vs. mean) must be scientifically justified. Random deletion is not allowed.

3. **Feature Engineering & Modeling**

- Feature Engineering: Create at least *one new feature* by merging or transforming >=3 existing columns. Explain its formula, the problem it solves, and prove it does not leak future information.
- Model Selection: Establish a simple Baseline model. Train at least 3 architecturally diverse models (e.g., Linear, Tree-based, Clustering/KNN).
- Failure Analysis (Mandatory): Execute and document an experiment that led to poor performance (e.g., inappropriate model or metric). Analyze the root cause scientifically and show the modification that improved it.

4. **Tuning & Evaluation**

- Hyperparameter Tuning: Select the best model and tune *exactly two (2)* hyperparameters via Cross-Validation. Explain their impact on Overfitting and Underfitting.
- Evaluation Metrics:
    - *Classification:* Use Confusion Matrix, Precision, Recall, F1-Score, and ROC-AUC. Select the most important metric based on the cost of error and justify it.
    - *Regression:* Use MAE, RMSE, and R^2. Explain outlier impact and plot actual vs. predicted values.
- Comparison: Provide a final comparison table justifying your ultimate model choice.

5. Business Insights & Submission Requirements

- Actionable Insights: Extract the top 3 predictive features (Feature Importance / SHAP) and provide 3 actionable business recommendations for the stakeholder.
- Code Quality Penalty: Code comments must explain the "*Why*" (decision rationale) not the "*What*". Descriptive-only comments will lose points.
- Deliverables:
    1. An organized, well-documented Jupyter Notebook.
    2. The saved final model (preferably a .pkl Pipeline).
    3. A copy of the Dataset (if licensing permits).
    4. A requirements.txt file.
    5. The Data Dictionary.
    6. A README file explaining reproduction steps.